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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2024

COMMON TO B.E – CSE / B.TECH – IT / AIDS & M.TECH - CSE

20MA214 – DISCRETE STRUCTURES AND COMBINATORICS

Answer ALL questions

Duration: 3 Hours

Maximum: 100 Marks

PART - A

(7x 1 = 7 Marks)

- CO2** If A has 4 elements and B has 3 elements then the number of functions from A to B is _____.
a) 3^4 b) 4^3 c) 16 d) 6
- CO2** The relation R on the set of all integers where $a R b$ if and only if $a = b^2$ is _____.
a) antisymmetric b) symmetric c) reflexive d) transitive
- CO1** Let N be the set of all natural numbers with ' $\leq$ ', defined as $m \leq n \Leftrightarrow n - m$, it is a non-negative integer for all m,n in N then $(N, \leq)$ is a _____.
a) poset and lattice b) poset but not lattice
c) neither poset nor lattice d) not poset but lattice
- CO3** Contra positive of $p \rightarrow q$ is _____.
a) $p \rightarrow \neg q$ b) $\neg p \rightarrow q$ c) $\neg p \rightarrow \neg q$ d) $\neg q \rightarrow \neg p$
- CO3** $P(x): x = x^3$ on the universe of integers the truth values
(i) $\exists x P(x)$
(ii) $\forall x P(x)$ are _____.
a) True (ii) - True b) True (ii)- False c) False (ii) -False d) False (ii)-True
- CO4** How many ways can 7 boys and 3 girls sit together in round table?
a) 6! b) 10! c) 3! d) 9!

7. **CO4** The number of ways to arrange the letters of the word “UNUSUAL” are _____.
 a) 800 b) 840 c) 7! d) 3!

Fill in the blanks:

(3x 1 = 3 Marks)

8. **CO1** If D is divisor, the Hasse diagram for $(D_{12}, /)$ is _____.
 9. **CO3** Express the negation of “If the teacher is absent, then some students do not keep quiet.” in symbolic form is _____.
 10. **CO4** If 15 dictionaries in a library contain a total number of 40,305 pages, then one of the dictionaries must have _____ pages by pigeonhole principle.

PART - B

(5x 3 = 15 Marks)

11. **CO1** Using characteristic function, prove that $f_{A \cup B}(x) = f_A(x) + f_B(x) - f_{A \cap B}(x)$ where A and B are subsets of universal set U , $x \in U$.
 12. **CO2** State and prove modular inequality on Lattice.
 13. **CO3** Find the disjunctive normal form of $(p \wedge \neg(q \wedge r)) \vee (p \rightarrow q)$
 14. **CO3** Without using truth tables, prove that $\neg(p \leftrightarrow q) \equiv (p \vee q) \wedge \neg(p \wedge q) \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$
 15. **CO4** How many bit string of length 10 contain (a) exactly four 1's (b) at least four 1's (c) an equal number of 0's and 1's?

PART - C

(5 x 15 = 75 Marks)

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any FOUR Questions

Compulsory Question:

16. **CO4** i) A Committee of 12 is to be selected from 10 men and 10 women. In how many ways can the selection be carried out if (8)
 • there are no restrictions?
 • there must be equal number of men and women?
 • there must be even number of women?
 • there must be more women than men?
 • there must be at least 8 men?

CO4 ii) Solve the recurrence relation (7)

$$a_{n+2} + 3a_{n+1} + 2a_n = 3^n, n \geq 0 \text{ given } a_0 = 0, a_1 = 1.$$

17. **CO3** i) Use rules of inference to prove that ‘Randy works hard, If Randy works hard then he is dull boy, and If Randy is a dull boy then he will not get the job imply the conclusion Randy will not get the job. (6)
- CO3** ii) Find the principal conjunctive normal form and principal disjunctive normal form of $(q \leftrightarrow p) \vee (\neg p \rightarrow r)$ without using truth table. (9)
18. **CO2** i) If $R = \{(1,2), (2,1), (2,2), (2,3), (3,1)\}$ and $S = \{(1,2), (2,2), (2,3), (3,1), (3,2), (3,3)\}$ are relations on a set $\{1,2,3\}$, find the matrix relations $M_R, M_S, M_{R \cup S}, M_{R \cap S}, M_{R \circ S}, M_{R \oplus S}, M_{S \circ R}$. (8)
- CO1** ii) If R is relation on Z defined by aRb if and only if $2a+3b=5n$ for some integer n , prove that R is an equivalence relation. (7)
19. **CO3** i) Using indirect method of proof, prove that (8)
- $$(a \rightarrow b) \wedge (a \rightarrow c), \neg(b \wedge c), (d \vee a) \Rightarrow d.$$
- CO1** ii) If S_n is the set of all divisors of the positive integer n and D is the relation of ‘division’ aRb if and only if a divides b , prove that $\{S_{24}, D\}$ is a Lattice (7)
- .Find also all the sub lattice of $D_{24} = \{S_{24}, D\}$ that contain 5 elements.
20. **CO1** i) If $f: Z \rightarrow N$ is defined by $f(x) = \begin{cases} 2x - 1 & x > 0 \\ -2x & x \leq 0 \end{cases}$ (8)
- Prove that f is one-to-one and onto and hence determine f^{-1} .
- CO3** ii) Show that the following premises “one student in this class knows how to write programs in JAVA” and ‘Everyone who knows how to write programs in JAVA can get a high paying job’ imply the conclusion “Someone in this class can get a high paying job “. (7)
21. **CO2** i) Prove that $f(x, y) = x * y$ is primitive recursive function. (8)
- CO4** ii) Assume that repetitions are not permitted, (7)
- How many four-digit numbers can be formed from the six digits 1,2,3,4,5,6?
 - How many four digit numbers are less than 4000?
 - How many even four-digit numbers can be formed from the six digits 1,2,3,4,5,6?

- How many four-digit odd numbers can be formed from the six digits 1,2,3,4,5,6?
- How many four-digit numbers can be formed from the six digits 1,2,3,4,5,6 are multiple of 5?
- How many four-digit numbers can be formed from the six digits 1,2,3,4,5,6 contain both the digits 3 and 5?

22. **CO4** i) Among 100 students 32 study Mathematics, 20 study statistics, 45 study accountancy, 15 study Mathematics and accountancy, 7 study Mathematics and statistics, 10 study statistics and accountancy and 5 studied all the three subjects. Find (i) the number of students studying none of these subjects? (ii) the number of students studying exactly one of the three subjects? (8)

CO2 ii) In any Boolean algebra, show that (7)

- $(a + b')(b + c')(c + a') = (b + a')(c + b')(a + c')$
- $ab' + a'b = 0$ if and only if $a = b$.
